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Data and Work Sheets - Print or bring it on an electronic device

Position, Velocity, and Acceleration - Physics 1401A
2026-2027

Please circle the appropriate values											
Course	**1101A**		**1201A**		**1401A**		**1501A**				
Lab Section	**002**	**003**	**004**	**005**	**006**	**007**	**008**	**009**	**010**	**013**	**014**
Lab Subsection	**A**	**B**	**C**	**D**							
Name	**First:** Beckett					**Last:** Graham					
Student #											
Lab Partner	**First:**					**Last:**					
Lab Station #											
Date	Sep 22, 2026										
Demonstrator	Brenna, Patriz										

Disclaimer: Please note that some but not all questions in this lab writeup will be graded.

INVESTIGATION 1: POSITION AND VELOCITY VS TIME GRAPHS

ACTIVITY 1: MOTION DETECTOR OPERATION

OBJECTIVES

- To familiarize yourself with the operation of the motion detector and *Logger Pro* software;
- To investigate position versus time graphs created by changing the speed and direction of a moving object.

APPARATUS

Logger Pro program for Windows, *LabQuest* interface, computer, motion sensor, meter ruler, student as the moving object, wooden board.

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PROCEDURE

If it is not already displayed, start the program by double clicking on *Activity 1*. Explore the *Graph Window* seen on the computer screen. You can change the scales on the axes by clicking on the first and last scale markers. Click on the last tick label on the horizontal *Time* scale, and change it from 10 to 20. Click on the last label on the vertical *Position* scale and change it from 3.5 to 5. Change them back to their original settings before you start your measurements.

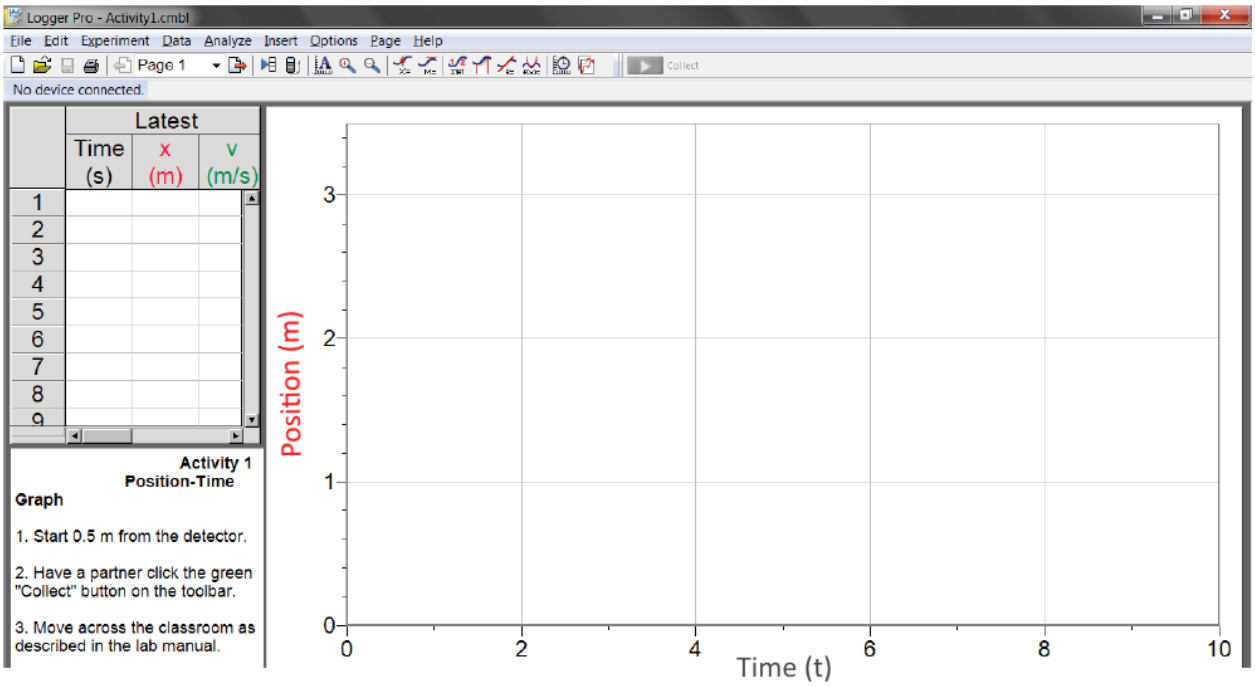


Figure 4: *Position-time* graph

Click once on the Collect button at the top of the screen to start the timer and data collection.

Make *position-time* graphs using one of your group members as the moving object. The object should use the wooden board as a reflector for the sound waves. It will take a bit of practice to get good data – hold the board steady and try to walk smoothly. Try collecting data while you are moving both *towards* and *away* from the detector. Walk or run at different speeds. Change directions! Make sure that you understand the resulting graphs. Every member of your group should take a turn.

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✓

Sketch a graph of position versus time obtained when you were walking slowly away from the sensor. You do not have to mark the scales or draw graphs to scale.

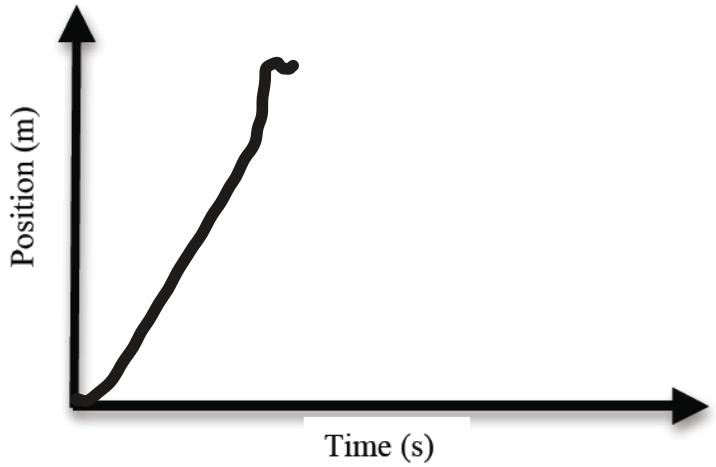


Figure 5: *Position-time* graph for walking away slowly.

Question 1: What aspect of the position-time graph changes when you change the speed of your motion?

the slope of the graph changes when you change the speed of motion

Question 2: What aspect of the position-time graph changes when you change the direction of your motion?

the sign of the slope changes when you change your direction of motion

Question 3: Probably you were unable to move in such a way that a graph of your position versus time was a perfectly straight line. Why?

the graph was imperfect because we could not stay at a constant speed while moving

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ACTIVITY 2: MATCHING A POSITION GRAPH

In this activity, you will try to move in a way that produces a position graph that matches a template provided.

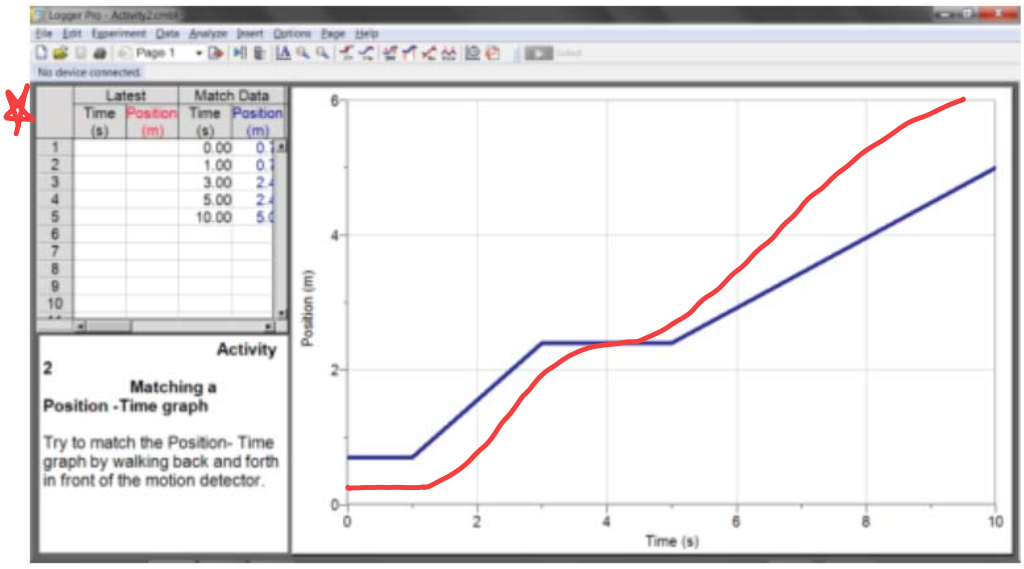


Figure 6: *Position vs time* graph to be matched.

Double click on *Activity 2* to open it. The graph displayed shows a plot of position versus time based on pre-defined data. Your task is to move in front of the detector so that you reproduce this graph as best you can.

Try a number of times, take turns, and work as a team. Sketch your best result on the Figure. ✓

Question 4: What parts of the graph were easy to reproduce, and what parts were difficult to reproduce. Why?

flat sections of the graph were easy because we had to stand still, slopes were hard to maintain constant velocity

Question 5: How did your motion differ on the two differently sloped parts of the graph?

originaly our motion was quicker and got slower in the second sloped part

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ACTIVITY 3: VELOCITY VERSUS TIME

INTRODUCTION

Velocity implies both *speed* and *direction*. How fast you move is your speed. Velocity is the rate of change of position with respect to time. In our experiment, it can be either positive or negative. If you move away from the detector, in the positive x direction, your velocity is positive, and if you move toward the detector, your velocity is negative.

METHOD

Open Activity 3 in *Logger Pro*.

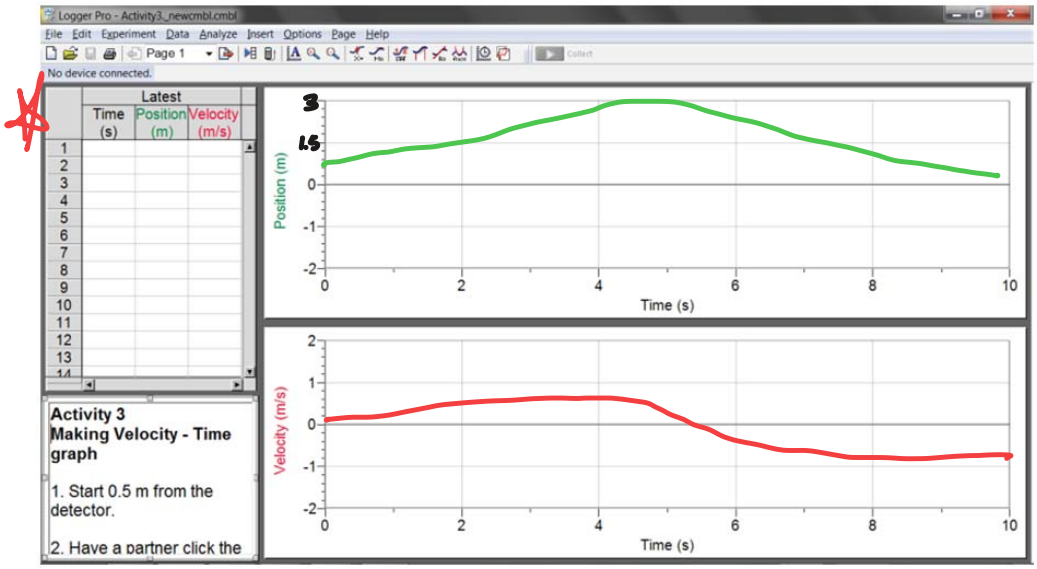


Figure 7: *Position* and *velocity vs. time* graphs.

Record a position-time graph while moving away from the detector at constant speed, just as you did above. Since position and velocity are related, the software can calculate your velocity using the equation $v = \Delta x/\Delta t$, where Δx is the change in position in the time interval Δt. It will automatically plot velocity versus time as it collects the data.

Estimate the slope of your position-time graph, and confirm that it is equal to the velocity plotted by the software. Repeat, but walking towards the detector. How does the velocity change?

Question 6: There is a regular pattern of bumps on your velocity-time graph. What causes these?

the bumps are caused by slight pauses in footsteps, aswell as changes in speed

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ACTIVITY 4: MATCHING A VELOCITY VERSUS TIME GRAPH

In this activity, you will try to reproduce a pre-defined velocity-time graph. Open *Activity 4* in *Logger Pro*. Move in front of the detector so that you reproduce the graph below as best you can. Try a number of times, and let everyone take a turn. Draw your group's best match on the plot in Figure 8.

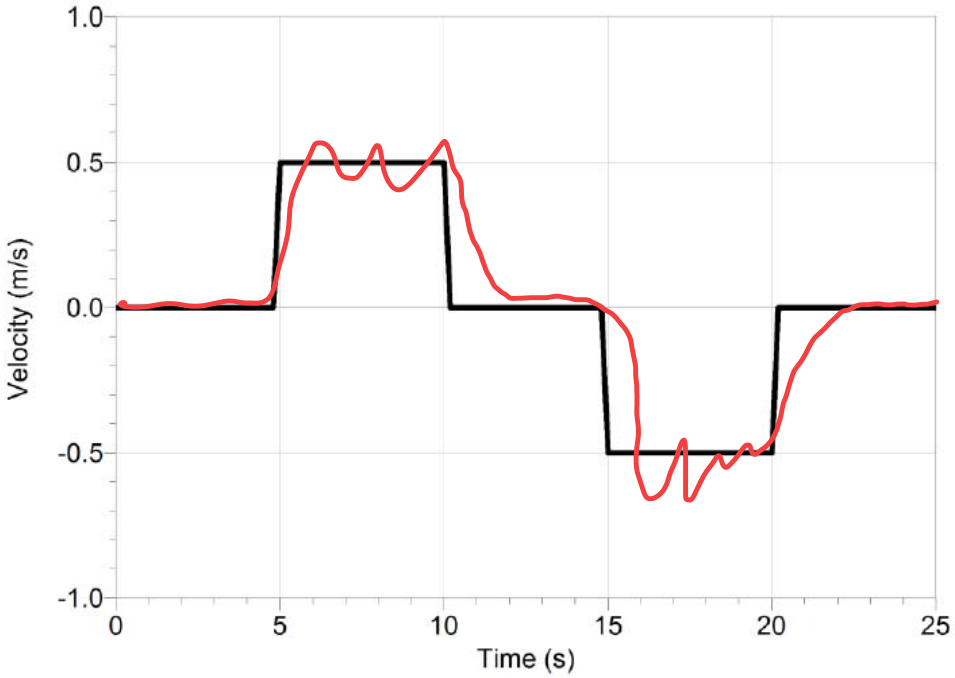


Figure 8: Matching a *velocity-time* graph.

Question 7: Which features of the graph were easiest to reproduce, and which were hardest? Explain why.

flat sections were easy because we remain still. the hardest parts were keeping at a constant velocity and switching directions fast and on time.

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INVESTIGATION 2: ACCELERATION - TIME GRAPHS OF YOUR MOTION

OBJECTIVE

To investigate acceleration versus time graphs and to study the relationship between position, velocity, and acceleration.

APPARATUS

Same as above.

INTRODUCTION

In this experiment, we have defined $x = 0$ to be at the detector, and x increases away from the detector.

Velocities away from the detector are positive, while velocities towards the detector are negative, as shown in Figure 9.

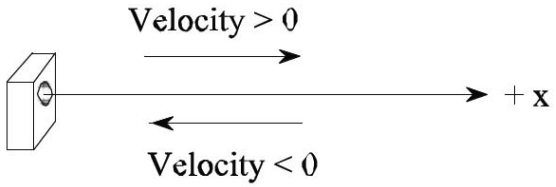


Figure 9: Sign of velocity

The acceleration is given by $a = \dfrac{v_2 - v_1}{t_2 - t_1}$

Example: Suppose the cart is moving **towards** the detector and is **slowing down**. Velocities v_1 and v_2 are both negative because the cart is moving towards the detector. Also, $|v_2| < |v_1|$ because the cart is slowing down. Suppose $v_1 = -2$ m/s at $t_1 = 0$ s and $v_2 = -1$ m/s at $t_2 = 10$ s. In this case:

$$a = \frac{(-1\text{m/s}) - (-2\text{m/s})}{10\text{ s} - 0\text{ s}} = +0.1\text{ m/s}^2 \tag{1}$$

We see that in this case a is positive.

Prediction: What is the sign of your acceleration if you **move away** from the detector

speeding up? positive

slowing down? Negative

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ACTIVITY 5: ACCELERATION FROM CONSTANT VELOCITY MOTION

Double click *Activity 5* to open it.

The window will display graphs of position, velocity, and acceleration. Starting at 0.5 m from the detector, make a position versus time graph by moving away from the detector at a constant speed. Probably your velocity-time graph is noisier than your position-time graph, and your acceleration–time graph is noisier still. This is because fluctuations in position (or velocity) are amplified when you subtract one data point from another to calculate velocity (or acceleration). During the period of time that your speed is constant, on average, your acceleration should be zero, on average. Is it?

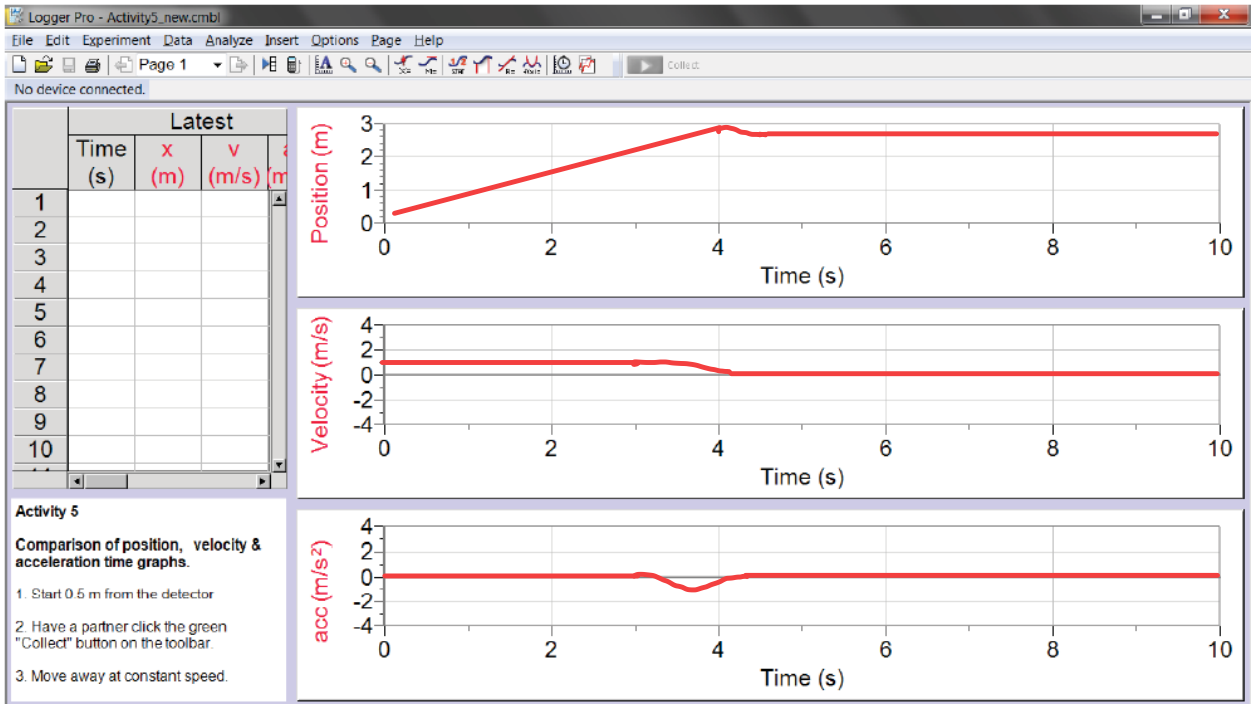


Figure 10: *Position*, *velocity* and *acceleration versus time* graphs.

ACTIVITY 6: ACCELERATION FROM MOTION WITH INCREASING VELOCITY

Double click *Activity 6* to open it.

Start at rest close to the detector, then move away with an increasing velocity – that is, start off slowly, and speed up as you go! Try it a few times, and take turns. Sketch your position, velocity, and acceleration data in Figure 11. Label the scale for the position, velocity and acceleration graphs. The time axis is common to all three graphs. Explain the velocity and acceleration graphs in terms of the slope of the position-time graph.

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DATA AND ANALYSIS

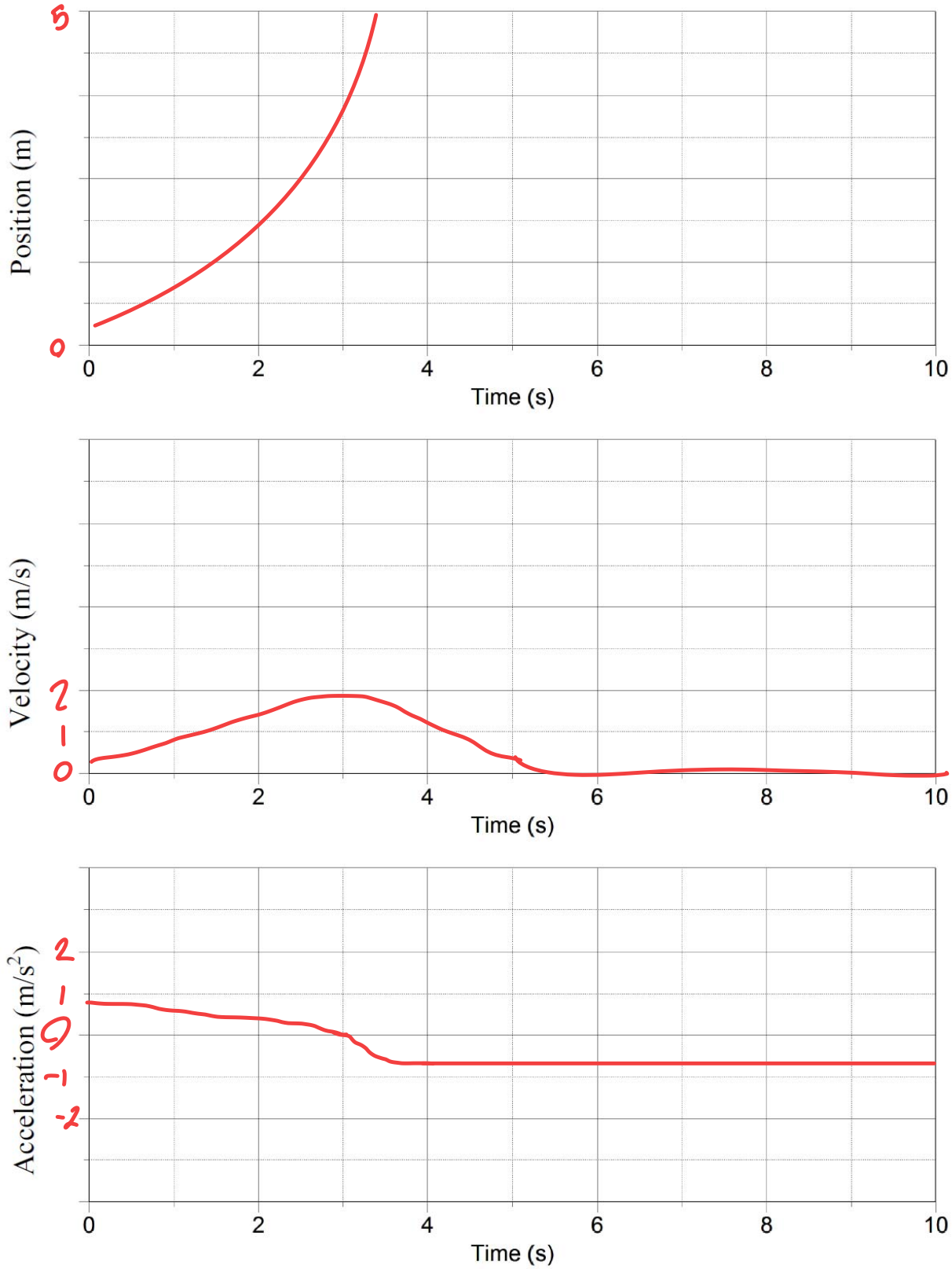


Figure 11: *Position*, *velocity* and *acceleration-time* graphs.

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Question 8: On average, was your acceleration positive or negative?

My average acceleration was positive because my velocity increased as I sped up

Now try it again, but this time start fast, then slow down. This is harder!

Question 9: During the time that you were slowing down, was your acceleration positive or negative on average?

My acceleration was negative on average while I was slowing down because my velocity was decreasing with time

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INVESTIGATION 3: CONSTANT ACCELERATION

OBJECTIVES

To investigate the motion of a cart on a track undergoing constant acceleration.

APPARATUS

Logger Pro 3.8.4 program for *Windows*, *LabQuest* interface, computer, motion sensor, low friction cart, track, pulleys, masses, string.

ACTIVITY 7: MOTION WITH CONSTANT ACCELERATION

METHOD

An easier way to study motion with constant acceleration is to use a cart on a track and accelerated by a constant force.

Set up the motion detector as shown in Figure 12. Open *Activity 7* and adjust the vertical position of the motion detector such that it detects the position of the cart along the entire track. Recall that the cart has to be a minimum of 15 cm from the sensor for it to be reliably detected. Attach the string to the cart, and run it over the pulley. Attach a small weight to the string so that it pulls the cart along the track. Experiment with different weights until you find one that pulls the cart the length of the track in a bit less than 10 s. Record the position versus time for your cart. Examine the graphs of velocity and acceleration. Adjust the axis scales as needed to display your data clearly.

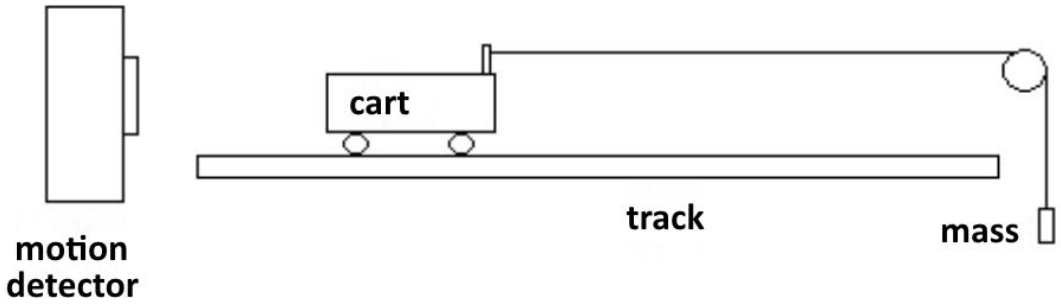


Figure 12: Cart on a track and pulled by a constant force, resulting in constant acceleration

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Sketch your position, velocity and acceleration versus time graphs in Figure 13.

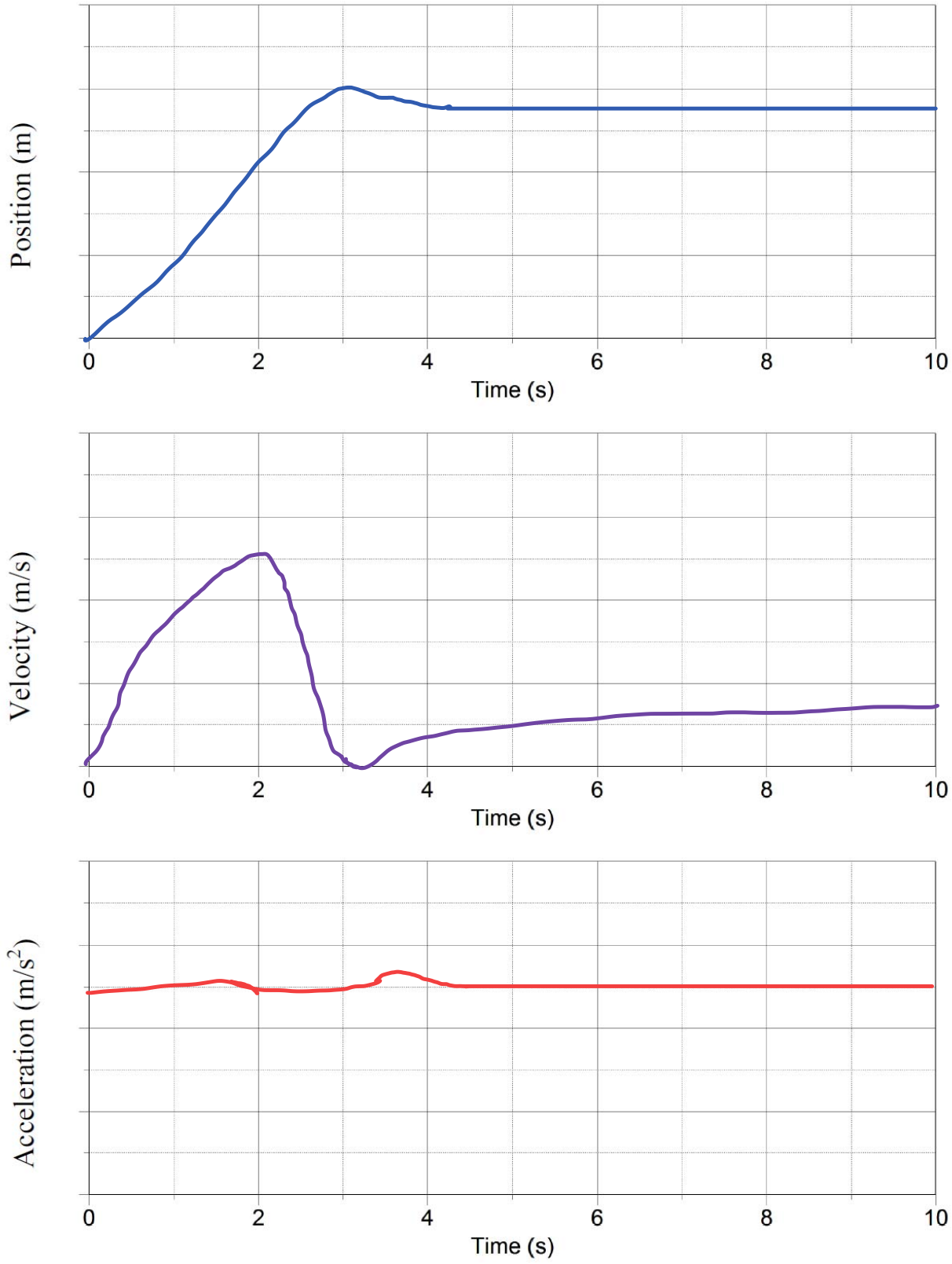


Figure 13: Motion of cart pulled by constant force.

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Question 10: Explain the motion of the cart based on the position, velocity, and acceleration graphs.

the cart is accelerating away from the sensor. the pT graph shows us its exponentialy growing possitive, that shows the velocity is increasing constantly. AT graph is a straight line showing positive acceleration

Question 11: Does your cart move with constant acceleration during any part of this experiment? When?

yes, the cart was moving with constant acceleration for the time after dropping the weight to the cart hitting the stopper

Question 12: Estimate the acceleration during this portion of the experiment from the slope of the velocity-time graph. Compare your result with the average value of the acceleration determined from your acceleration-time graph. You don't need to be really precise here; a rough comparison is all that is required.

$$M = \frac{\Delta V}{\Delta t} = \frac{0.5 - 0.2}{2 - 1} = 0.3\ m/s^2$$

Using the VT graph we tooks two points and found the avg acceleration which matches the same time point on AT graph. from that data we found acceeleration of $0.3 m/s^2$

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CONCLUSIONS AND DISCUSSION

Connect the data from your experiments with the learning outcomes specified at the beginning of the lab manual.

Final Mark	